

DOCUMENT 1: INITIAL NETWORK DIAGRAM SPECIFICATION

Project Component: Task 4: Initial Network Diagram

1.0 Executive Topology Summary

The logical network layout for Ubuntu Innovations (Pty) Ltd implements a rigid, high-performance hierarchical design tailored to support 25 localized enterprise users. Data routing moves systematically along a North-to-South path. This design ensures that all internal Local Area Network (LAN) nodes are shielded from unauthenticated public traffic while maintaining high-speed access to central network assets and external cloud systems.

2.0 Network Asset & Media Path Blueprint

- **The Internet Edge Link**
 - A high-speed fiber uplink provides primary broadband access. The physical media line runs directly from the Internet Service Provider (ISP) demarcation point into the public-facing Wide Area Network (WAN) port of the security gateway.
- **Perimeter Security & Core Routing Layer**
 - The **Integrated Edge Router & Firewall** acts as the definitive security gateway. It processes external Network Address Translation (NAT), stateful inspection packet filtering, secure access lists, and dynamic host configuration protocol (DHCP) subnet scope allocations.
- **Core Network Distribution Layer**
 - A dedicated Cat6 copper Ethernet backbone cable links the firewall's internal LAN interface to the primary **24/48-Port Core Managed Network Switch**. This appliance acts as the central switching core for all internal local area data distributions.
- **Wired Core Node Assignments (LAN)**
 - Dedicated, hardwired Cat6 copper Ethernet patch cables drop from the managed switch patch panel directly into local physical endpoints to completely mitigate packet loss and signal attenuation:

- **Fixed Workstations (Desktops):** Distributed across employee desk pools to guarantee low-latency, stable access to productivity software.
- **Network-Attached Storage (NAS) Backup Device:** Wired via a high-speed interface directly to the switch core to secure automated backup data flows and internal file shares.
- **Shared Network Printers:** Configured with fixed, static IP addresses to remain continuously resolvable across separate office departments.
- **Wireless Infrastructure Layer (WLAN)**
 - Distributed **Wireless Access Points (WAPs)** connect directly into the core managed switch using **Power over Ethernet (PoE)**. This mechanism delivers both data connectivity and electrical power through a single Cat6 line run. These access points broadcast an encrypted Wi-Fi spectrum secured via WPA3 protocols to safely onboard mobile staff laptops and smartphone endpoints.

